Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Cardoso Dairy

Physical address of dairy:

2720 Kenney AVE	Winton	Merced	95388
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/05/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X052-X020-X033-XXXX

B. OPERATORS

Cardoso, Antonio & Guiomar

Operator name: Cardoso, Antonio & Guiomar

Telephone no.: Landline: ; Cellular: (209) 678-1176

2720 Kenney AVE	Winton	CA	95388
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Cardoso, Antonio & Guiomar

Legal owner name: Cardoso, Antonio & Guiomar

Telephone no.: Landline: ; Cellular: (209) 678-1176

2720 Kenney AVE	Winton	CA	95388
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	150	150	0
Number under roof	300	45	300	0	0	100
Maximum number	300	45	300	150	150	100
Average number	300	45	300	150	150	100
Avg live weight (lbs)	1,000	1,100	750	500		

Predominant milk cow breed: Jersey

Average milk production: 52 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 12,730.29 *tons per reporting period*

Total nitrogen from manure: 156,020.45 *lbs per reporting period*

After ammonia losses (30% loss applied): 109,214.32 *lbs per reporting period*

Total phosphorus from manure: 24,724.19 *lbs per reporting period*

Total potassium from manure: 53,904.97 *lbs per reporting period*

Total salt from manure: 151,602.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,328,000 *gallons*

Total nitrogen generated: 29,440.56 *lbs*

Total phosphorus generated: 3,246.94 *lbs*

Total potassium generated: 46,793.42 *lbs*

Total salt generated: 313,782.68 *lbs*

	5,328,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	5,328,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal (Merced Irr. District)	Surface water
Dairy/Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	Scraped material	1,250.00 *ton*	Dry-weight	20.9		24,400.00	6,950.00	15,400.00		0.00
12/15/2023	Corral solids	875.00 *ton*	Dry-weight	13.7		30,000.00	7,350.00	22,800.00		41.90

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	93,558.50	24,843.96	64,887.20	632,794.75
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	93,558.50	24,843.96	64,887.20	632,794.75

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
#1	62	62	4	process wastewater	X052-X020-X033-XXXX
Totals for areas that were used for application	62	62	4		
Totals for areas that were not used for application					
Land application area totals	62	62	4		

B. CROPS AND HARVESTS

#1

Field name: #1

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 62 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	600.00 *ton*	Dry-weight		70.4	14,700.00	2,250.00	20,000.00		9.00
04/30/2023	330.00 *ton*	Dry-weight		69.0	13,000.00	2,260.00	20,600.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	139.50	0.00
Total actual harvest content	15.00	127.12	20.35	182.56	852.21

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 62 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	784.00 *ton*	Dry-weight		75.2	15,700.00	1,730.00	9,700.00		5.90
09/27/2023	1,280.00 *ton*	Dry-weight		73.0	17,700.00	2,320.00	12,400.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	304.00	41.60	214.40	0.00
Total actual harvest content	33.29	295.80	36.71	199.08	1,083.54

NUTRIENT BUDGET

A. LAND APPLICATIONS

#1 - 10/28/2022: Oats, silage-soft dough

Field name: #1

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	109.72	13.43	158.16	1,304.79	1,224,000.00 *gal*
Application event totals		109.72	13.43	158.16	1,304.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.65	6.15	94.20	593.09	648,000.00 *gal*
Application event totals		57.65	6.15	94.20	593.09	

#1 - 05/25/2023: Corn, silage

Field name: #1

Crop: Corn, silage Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	153.74	16.40	251.19	1,581.57	1,728,000.00 *gal*
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	7,020,000.00 *gal*
Application event totals		153.74	16.40	251.19	1,581.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	6,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

#1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	6,255,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	76.87	8.20	125.59	790.78	864,000.00 *gal*
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	6,255,000.00 *gal*
Application event totals		76.87	8.20	125.59	790.78	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	6,255,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	6,255,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	76.87	8.20	125.59	790.78	864,000.00 *gal*
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	6,255,000.00 *gal*
Application event totals		76.87	8.20	125.59	790.78	

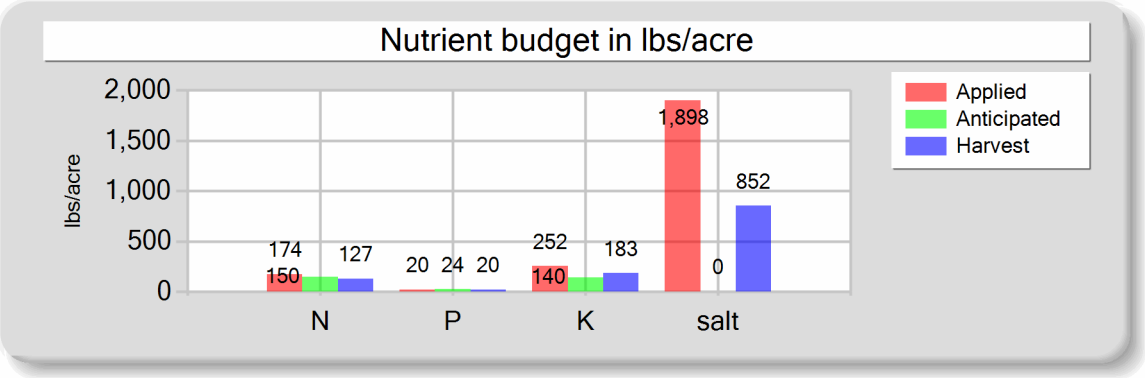
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal (Merced Irr. District)	Surface water	0.00	0.00	0.00	0.00	6,255,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

B. NUTRIENT BUDGET

#1 - 10/28/2022: Oats, silage-soft dough

Field name: #1 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	167.37	19.58	252.35	1,897.88
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.37	19.58	252.35	1,897.88
Anticipated crop nutrient removal	150.00	24.00	139.50	0.00
Actual crop nutrient removal	127.12	20.35	182.56	852.21
Nutrient balance	47.26	-0.77	69.79	1,045.67
Applied to removed ratio	1.37	0.96	1.38	2.23

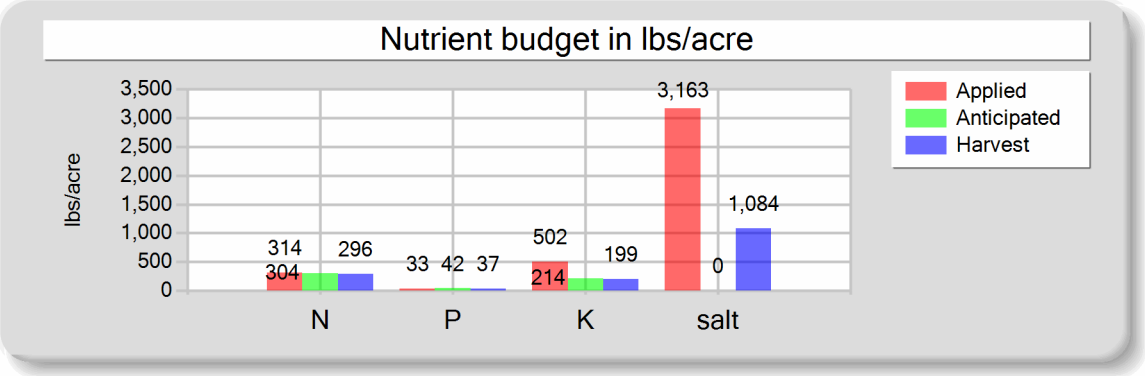
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,872,000.00 *gallons*
68.94 *acre-inches*
1.11 *inches/acre*

Total harvests for the crop
2 *harvests*

#1 - 05/25/2023: Corn, silage

Field name: #1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	307.48	32.79	502.38	3,163.13
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.48	32.79	502.38	3,163.13
Anticipated crop nutrient removal	304.00	41.60	214.40	0.00
Actual crop nutrient removal	295.80	36.71	199.08	1,083.54
Nutrient balance	18.68	-3.92	303.30	2,079.59
Applied to removed ratio	1.06	0.89	2.52	2.92

Fresh water applied	
51,120,000.00	*gallons*
1,882.58	*acre-inches*
30.36	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
2.05	*inches/acre*

Total harvests for the crop	
2	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T3306005G

Sample and source description: T3306005G

Sample date: 11/02/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,000.00	7,350.00	22,800.00	17,500.00	10,500.00	5,670.00	5,240.00	1,320.00		41.90
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3346026A

Sample and source description: T3346026A

Sample date: 12/12/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 20.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	6,950.00	15,400.00							0.00
DL	1.00	1.00	1.00							1.00

B. PROCESS WASTEWATER ANALYSES

T2362001B

Sample and source description: T2362001B

Sample date: 12/28/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	666.00	347.00	0.00	0.00	81.50	960.00								7,130.00	7,920
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3101042A

Sample and source description: T3101042A

Sample date: 04/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	661.00	414.00	0.00	0.00	70.50	1,080.00								7,910.00	6,800
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3306005B

Sample and source description: T3306005B

Sample date: 11/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	540.00	386.00	0.00	0.00	46.00	704.00								7,120.00	5,540
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

Canal (Merced Irr. District)

T3306005A

Sample description: T3306005A

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.14	0.50	0.14								100.00	1

Dairy/Domestic Well

Dairy/Domestic Well

T3089013A

Sample description: T3089013A

Sample date: 03/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			1.50									
DL			1.00									

T3346026B

Sample description: T3346026B

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.13	0.00	1.13	11.00	3.62	9.49	62.00	0.00	3.28	2.17	158.00	105
DL	1.00	0.50	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

#1 - 10/28/2022: Oats, silage-soft dough

T3306005C

Sample and source description: T3306005C

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	2,250.00	20,000.00		9.00
DL	1.00	1.00	1.00		1.00

#1 - 10/28/2022: Oats, silage-soft dough

T3306005D

Sample and source description: T3306005D

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,260.00	20,600.00		10.20
DL	1.00	1.00	1.00		1.00

#1 - 05/25/2023: Corn, silage

T3306005E

Sample and source description: T3306005E

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	1,730.00	9,700.00		5.90
DL	1.00	1.00	1.00		1.00

T3306005F

Sample and source description: T3306005F

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	2,320.00	12,400.00		6.40
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

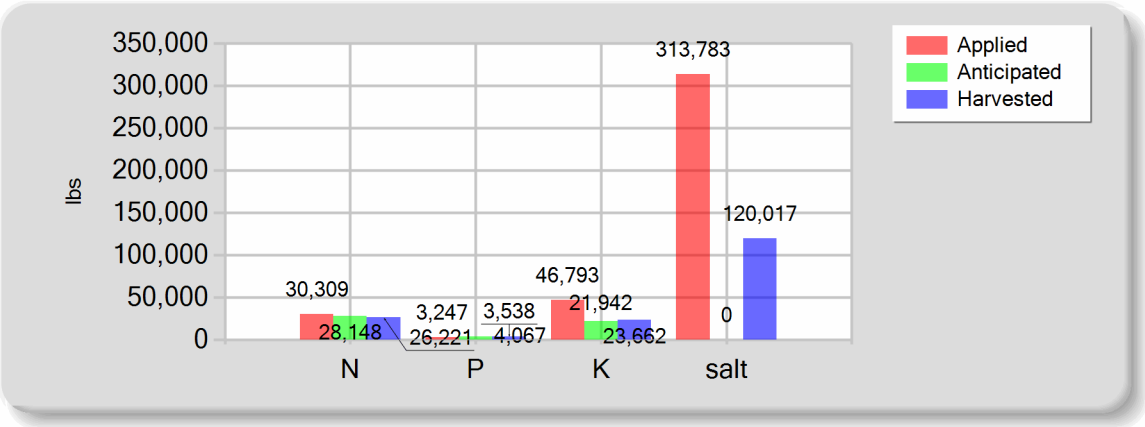
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

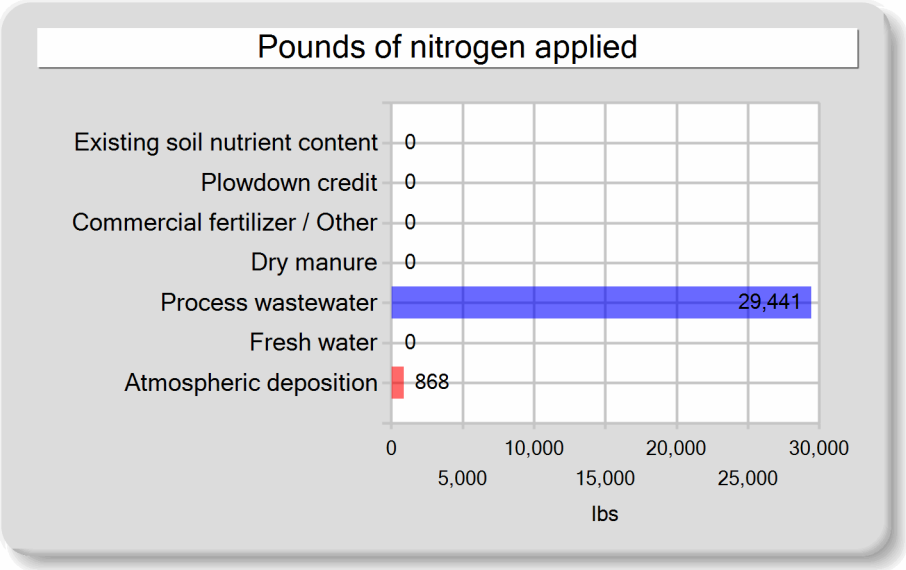
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

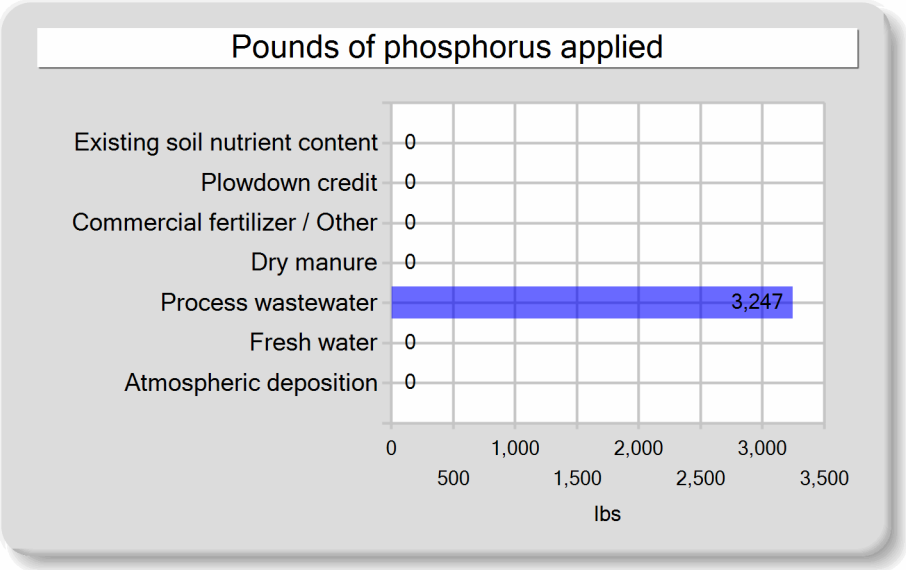
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	29,440.56	3,246.94	46,793.42	313,782.68
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	868.00	0.00	0.00	0.00
Total nutrients applied	30,308.56	3,246.94	46,793.42	313,782.68
Anticipated crop nutrient removal	28,148.00	4,067.20	21,941.80	0.00
Actual crop nutrient removal	26,220.64	3,537.91	23,661.62	120,016.98
Nutrient balance	4,087.91	-290.98	23,131.80	193,765.71
Applied to removed ratio	1.16	0.92	1.98	2.61

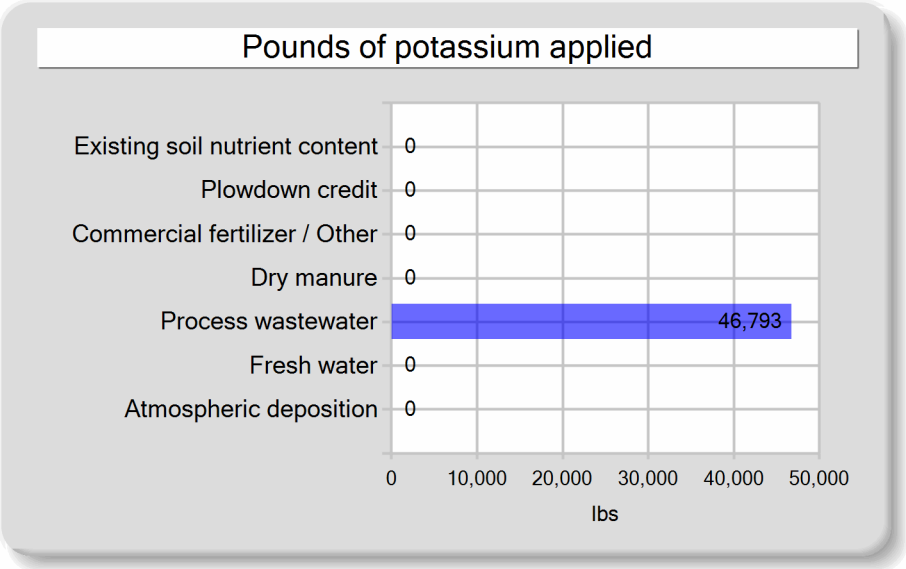
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

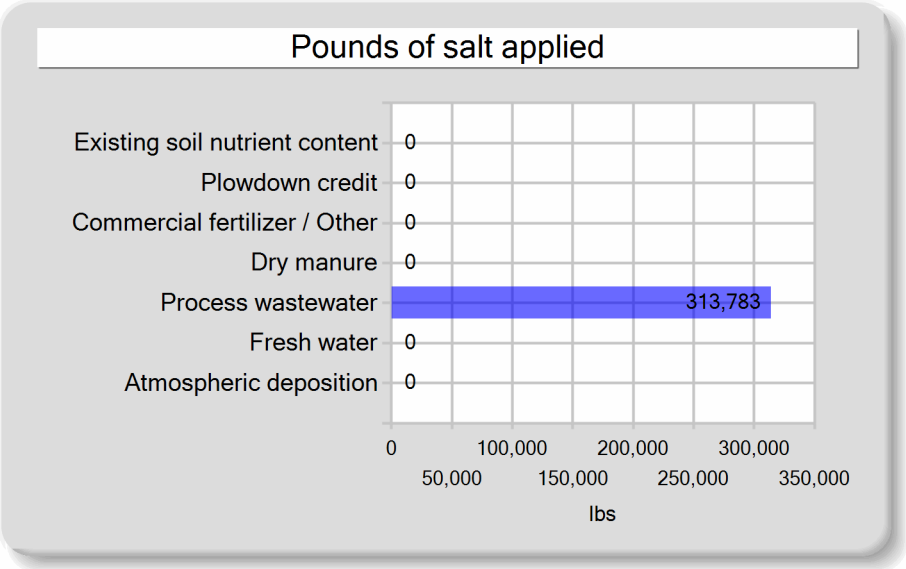


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Antonio Cardoso
SIGNATURE OF OWNER OF FACILITY

Antonio & Guiomar Cardoso
PRINT OR TYPE NAME

6/12/24
DATE

SIGNATURE OF OPERATOR OF FACILITY

SAME AS OWNER
PRINT OR TYPE NAME

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Antonio Cardoso_____

Name of Dairy Facility: __Cardoso Dairy_____

Facility Address: __2720 Kenney Ave._____Winton_____95388__
Number and Street — City — Zip Code

Contact Person Name and Phone Number:_Guiomar Cardoso_____(209) 678-1176_
Name — Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: __Stone Spreading_____

Address of Hauling Company /Person:__5175 Shaw Ave._____Winton_____95388_
Number and Street — City — Zip Code

Contact Person:__Noel Perez_____(209) 675-1470_____
Name — Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_Stone Spreading____5175 Shaw Ave._____Winton_____95388_____(209) 675-1470_
Name — Number and Street — City — Zip Code — Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

___5175 Shaw Ave._____Winton_____95388_____
Number and Street — City — Zip Code — Assessor's Parcel Number

Dates Hauled: __10/30 – 12/15/23_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 3.0 % P 0.7351 % K 2.28 %

Manure: 875____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _41.9 %_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of Manure: Hauler's Estimation

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes ___X___ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
___N/A_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Antonio Cardoso Date: 6/14/24

Hauler's Signature: Noe Perez Date: 6/14/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ___Antonio Cardoso_____

Name of Dairy Facility: _Cardoso Dairy_____

Facility Address: _2720 Kenney Ave._____Winton_____95388__
Number and Street City Zip Code

Contact Person Name and Phone Number:_Guiomar Cardoso_____(209) 678-1176_
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Michael Bettencourt_____

Address of Hauling Company /Person:_11584 N. Schaffer Rd._____Winton_____95388___
Number and Street City Zip Code

Contact Person:_Michael Bettencourt_____(209) 617-2255_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_____SAME AS HAULER_____
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____SAME_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: _October 1 – November 15, 2023 _____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 1.24% P 0.343% K 1.68%

Manure: _1250_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): __29.6_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: hauler's estimate

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Antonio Castro Date: 3/13/24

Hauler's Signature: Michael A Pettimore Date: 3/13/24

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Antonio Cardoso Dairy
2720 Kenney Avenue
Winton, CA 95388

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER: Merced County
RECEIVED DATE: 3/30/23 3:18 PM
SUBMITTED BY:
ANALYZED DATE: 3/31/23

Water Potability Test - P/A (WPT-P/A)
SOURCE: Clear Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Water Sample	T3089013A	Nitrate Nitrogen (NO3-N)	EPA 353.2	1.50	mg/L
		Total Coliform (TC)	SM 9223 B (Colilert)	Absent	CFU/100mL
		Escherichia Coli (E Coli)	SM 9223 B (Colilert)	Absent	CFU/100mL

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Antonio Cardoso Dairy
2720 Kenney Avenue
Winton, CA 95388

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:16 PM
SUBMITTED BY:
ANALYZED DATE: 12/15/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Water D.W.	T3346026B	Electrical Conductivity (EC)	EPA 120.1 and	0.158	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	62.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	3.28	mg/L
		Sodium (Na)	EPA 200.7	9.49	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	3.62	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.13	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	105	mg/L
		Calcium (Ca)	EPA 200.7	11.0	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	2.17	mg/L
		Soluble Salts (SALT-SOL)	Calculation	101	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)			1.81
Calcium (Ca)	101			178 (T3346011A)	171 (DUP)	3.72
Chloride (Cl)	106	107 (T3346026B)	104 (T3346026B)			2.47
Magnesium (Mg)	106			137 (T3346011A)	137 (DUP)	0.285
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)			1.40
Sodium (Na)	97.0			548 (T3346011A)	539 (DUP)	1.51
Sulfate (SO4)	102	95.9 (T3346026B)	95.1 (T3346026B)			0.864
Total Dissolved Solids (TDS)	101			880 (W3346003B)	860 (Dup)	2.30

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

WATER SAMPLE SUBMITTAL FORM

DENELE ANALYTICAL INC.
865 S. Kilroy Rd. Turlock CA, 95380 (209) 634-9055
40 North E Street, Ste. B Woodland, CA 95776 (530) 666-9056
info@denelelabs.com

Report & Bill To:	
Company:	Antonio Cardoso
Address:	
City, State & Zip	
Email to:	
Submitted By:	Guiomar Cardoso
Grower (if applicable):	

Date Collected:	Time Collected:	Sample ID:	Complete (COM-W)*	Phosphorus (P)	Sulfate (SO4)	Zn, Fe, Cu, Mn	pH only	NO3-N	E. coli P/A or #	Tot. Coliform P/A or #	WPT*	Other - Please List
3/30/23	2:10	Water Sample									X	21.2°c

***Complete:** pH, EC, Carbonates, Bicarbonates, NO3, NO3-N, Soluble Salts, Ca, Mg, Na, K, B, Chloride, Sulfate, & Sodium Absorption Ratio (SAR)

***WPT (Water Potability):** NO3-N, Total Coliform P/A, & E. coli P/A

Are analysis requested for regulatory purposes? (Only applicable to NO3-N, E. coli, & Coliform) (Y/N): Y Merced County

If Y, please list regulatory agency and sign below:

Signature	Company	Received Date & Time	Relinquished Date & Time	LAB USE ONLY
Guiomar Cardoso	3:18		3:18	LABORATORY NUMBER
	DENELE ANALYTICAL INC.	3-30-23 3:18		T3089013

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Antonio Cardoso Dairy
2720 Kenney Avenue
Winton, CA 95388

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/2/23 12:04 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/3/23

Irrigation/ Ground Water (H1)
SOURCE: Canal Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Canal	T3306005A	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	16.5	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6	97.6 (T3306002A)	95.2 (T3306002A)	2.48
Electrical Conductivity (EC)	98.9			0.297
Nitrate Nitrogen (NO3-N)	99.3	103 (T3306005A)	101 (T3306005A)	2.39

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Antonio Cardoso Dairy
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: KCPK Inc.-(Jackie)
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/2/23	10am	Canal		21.0°c		X				
2	11/2/23	10am	Lagoon		23.0°c			X			
3	11/2/23	10am	Oat 1							X	
4	11/2/23	10am	Oat 2							X	
5	11/2/23	10am	Corn 1							X	
6	11/2/23	10am	Corn 2							X	
7	11/2/23	10am	Corral manure								M7
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Guillermo Cardoso					
2nd	[signature]	DENELE ANALYTICAL, INC.	11/2/23 12:04			T33060005

Required to be submitted to any agency? (YES / NO)